Agenda

1.) Announcements:

- Exam 1 is next week
- 7.8 and 9.1 HW due Wednesday

2.) Section 9.2

3.) Start Section 9.3

Definition: A direction field (or slope field) of a differential equation is a graph of short slope lines that correspond to the differential activity at a particular point.

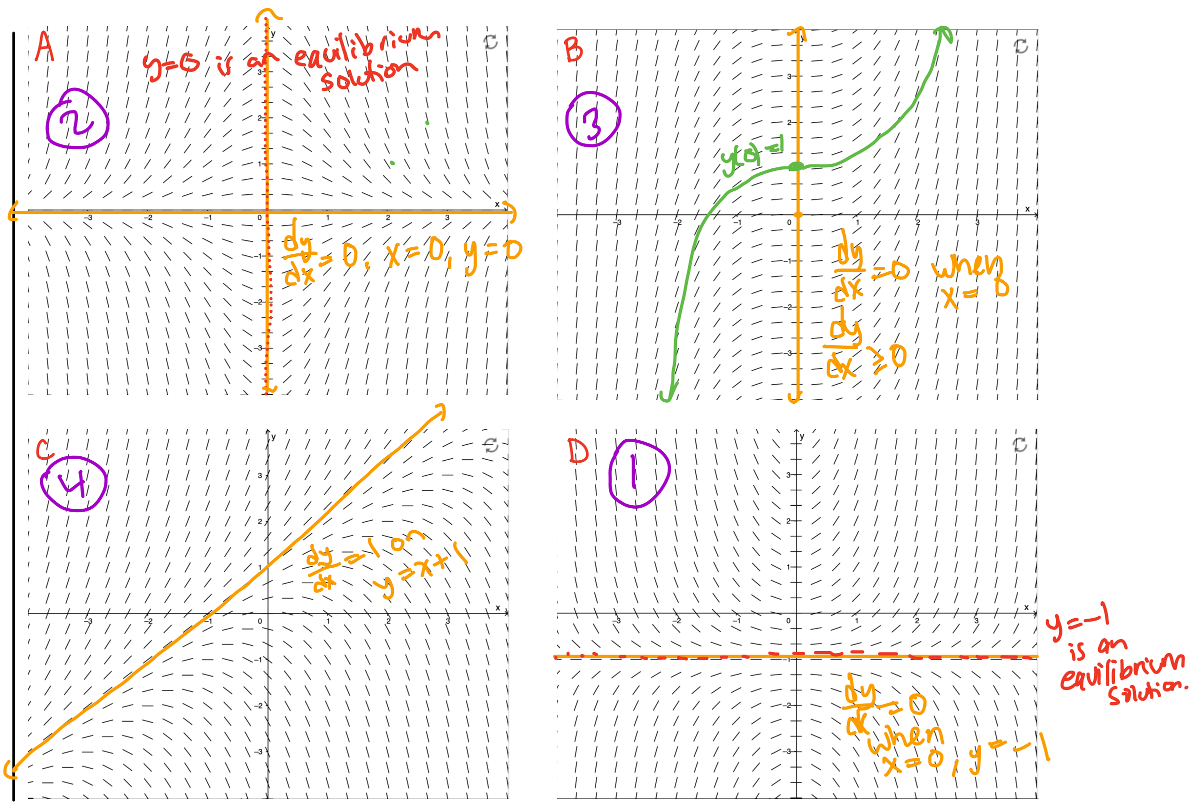
Example: Match the differential equation to the direction field

1.) $\frac{dy}{dx} = x + xy$
$= x(1+y)$

2.) $\frac{dy}{dx} = -xy$

3.) $\frac{dy}{dx} = x^2$

4.) $\frac{dy}{dx} = y - x$
$= y + -x$



Euler's Method

We can find approximate solutions to differential equations using a numeric method that relies on linear approximations.

Example: Let $\overbrace{\frac{dy}{dx} = \frac{x}{y}}^{\frac{dy}{dx} = F(x,y)}$ Given $\overbrace{y(0)=1}^{\text{initial condition}}$, approximate $\overbrace{y(2)}^{\text{what we want to approx.}}$

using a step size of $\underbrace{h = 0.5}$, "hops" we make from start to finish

i. $(x_0, y_0) = (0,1)$

ii. $\left.\frac{dy}{dx}\right|_{(0,1)} = F(0,1) = \frac{0}{1} = 0$

$(x_1, y_1) = (\frac{1}{2}, \underline{1 + \overset{h}{(\frac{1}{2})}\overset{F(0,1)}{(0)}}) = (\frac{1}{2}, 1)$

iii. $\left.\frac{dy}{dx}\right|_{(\frac{1}{2},1)} = F(\frac{1}{2},1) = \frac{1/2}{1} = \frac{1}{2}$

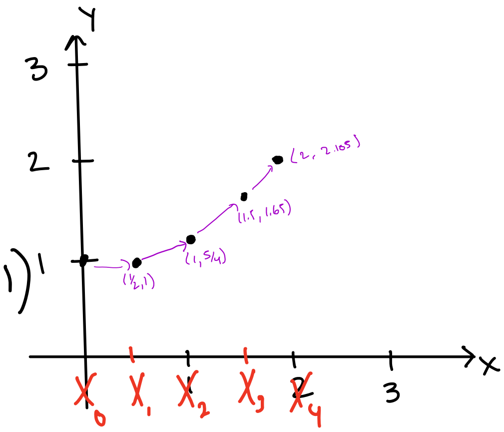
$(x_2, y_2) = (1, \underline{1 + \overset{h}{(\frac{1}{2})}\overset{F(\frac{1}{2},1)}{(\frac{1}{2})}}) = (1, \frac{5}{4})$

iv. $\left.\frac{dy}{dx}\right|_{(1,\frac{5}{4})} = F(1, \frac{5}{4}) = \frac{4}{5}$

$(x_3, y_3) = (1.5, \underline{1.25 + \underset{h}{(.5)}\underset{m}{(.8)}}) = (1.5, 1.65)$

v. $\left.\frac{dy}{dx}\right|_{(1.5,1.65)} = F(1.5, 1.65) \approx .909$

$(x_4, y_4) = F(2, 1.5 + (.5)(.909)) \approx (2, 2.105)$



slope $= \frac{\Delta y}{\Delta x}$

$F(0,1) = \frac{\Delta y}{h}$

$hF(0,1) = \Delta y$

Euler's Method Let $\frac{dy}{dx} = F(\overset{}{x}, y)$ $\; x/y$. Suppose we have an initial-value $y(\overset{0}{x_0}) = \overset{1}{y_0}$. Given a step size $\overset{1/2}{h}$, at $x_n = x_{n-1} + h$, we have $\underline{y_n = y_{n-1} + hF(x_{n-1}, y_{n-1})}$,

$n = 1, 2, 3, \ldots$

Activity: Let $y' = 1 - xy$. Given $y(0) = -1$, use Euler's method with $h = 0.5$ to estimate $y(1)$.

~~1~~ 8 ~~10~~ ~~19~~ ~~21~~ 22 ~~35~~ ~~39~~

~~opp. big toes~~
~~thin enamel~~
~~Minor dimorphism~~
long, hook-like fingers

Sociality
arborealism
omnivory
tropical distribution

class

instance vars

static

Constructor

getters
setters
mutators

main {

Section 9.3 - Separable Equations

A separable equation is a first-order differential equation that can be expressed as $\frac{dy}{dx} = g(x)f(y)$ where $g(x)$ is a function of x and $f(y)$ is a function of y.

Activity: Which of the following are separable?

1. $\frac{dy}{dx} = xy^2$

2. $\frac{dy}{dt} = \frac{8t^3 + 2t}{y^2}$

3. $\frac{dy}{dx} = e^{7x} + 5y$

4. $\frac{dy}{dt} = \frac{t}{t^2y + y}$

A first-order equation is autonomous if it can be expressed as $\frac{dy}{dx} = f(y)$ where $f(y)$ is a function of y.